Program: **Diploma in Food Processing Technology**	
Course Code: **5427**	Course Title: **Unit operations of food industry Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Program core**	
Periods per week: **3(L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- Identify various experiments for identification of properties of fluids
- Understand various techniques for determining thermal properties, heat transfer, mass transfer

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Awareness about unit operations in food industry		unit operations in food industry	V

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Determine the coefficient of discharge and calibrate both Venturi meter and Orifice meter.	11	Applying
CO2	Apply the principles of heat transfer to determine the rate of heat transfer,heat transfer coefficient, and efficiencies in various heat exchanger systems and steam distillation processes.	14	Applying
CO3	Apply principles to determine the rate of drying and plot drying curves for various dryers, and calculate minimum fluidization and terminal settling velocities in fluidized bed and rotary dryers.	9	Applying

CO4	Apply the principles of filtration and leaching to determine the rate of filtration for constant pressure filtration and the overall stage efficiency of a continuous cross- current leaching unit for extracting sodium carbonate	11	Applying

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3				\		

3-Stronglymapped, 2-Moderatelymapped, 1-Weaklymapped

Course Outline

Module Outcomes	Descrip tion	Duration (Hours)	Blooms Taxonomy Level
CO1	**Determine the coefficient of discharge and calibrate both Venturi meter and Orifice meter.**		
M1.01	Determination of coefficient of discharge of Venturi meter	2	Understanding
M1.02	To Calibrate a venturi meter	3	Applying
M1.03	Determination of coefficient of discharge of Orifice meter	3	Applying
M1.04	To Calibrate a orifice meter	3	Applying
CO2	**Apply the principles of heat transfer to determine the rate of heat transfer, heat transfer coefficient, and efficiencies in various heat exchanger systems and steam distillation processes.**		
M2.01	Determination of the rate of heat transfer and heat transfer coefficient of Shell and tube heat exchanger	6	Applying
M2.02	Determination of thermal and vapourization efficiencies for steam distillation of turpentine	3	Applying

M2.03	Experiment on "Heat transfer in double pipe heat exchanger in laminar flow .	3	Applying
	Lab Test –I	2	
CO3	**Apply principles to determine the rate of drying and plot drying curves for various dryers, and calculate minimum fluidization and terminal settling velocities in fluidized bed and rotary dryers.**		
M3.01	Determination of rate of drying and plotting drying curve of natural circulation dryer	3	Applying
M3.02	Determination of the minimum fluidization velocity and terminal settling velocity of Fluidised bed dryer	3	Applying
M3.03	Determination of the rate of drying for the given food grains in Rotary dryer	3	Applying
CO4	**Apply the principles of filtration and leaching to determine the rate of filtration for constant pressure filtration and the overall stage efficiency of a continuous cross- current leaching unit for extracting sodium carbonate**		
M4.01	Determination of rate of filtration for constant pressure filtration	3	Applying
M4.02	Determination of the overall stage efficiency of a continuous cross-current leaching unit for extracting sodium carbonate from a mixture of sodium carbonate and sand by conducting a batch laboratory experiment	6	Applying
	Lab Test– II	2	

Text/Reference

T/R	**Book Title/Author**
R1	Richardson, J.E. etal.,"Coulson & Richardson's Chemical Engineering" Vol.2 (Praticle Technology & Separation Processes") 5th Edition, Butterworth – Heinemann / Elsevier, 2003.

Online Resources

Sl.No	Website Link
1.	https://testbook.com/mechanical-engineering/venturimeter-definition-properties-and-types
2.	https://testbook.com/mechanical-engineering/orifice-meter-formula-diagram-and-working-principle
3.	https://webbook.nist.gov/chemistry/fluid/